



Compensatory load redistribution in walking and trotting dogs with hind limb lameness



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ABSTRACT

This study evaluated adaptations in vertical force and temporal gait parameters to hind limb lameness in walking and trotting dogs. Eight clinically normal adult Beagles were allowed to ambulate on an instrumented treadmill at their preferred speed while the ground reaction forces were recorded for all limbs before and after a moderate, reversible, hind limb lameness was induced. At both gaits, vertical force was decreased in the ipsilateral and increased in the contralateral hind limb. While peak force increased in the ipsilateral forelimb, no changes were observed for mean force and impulse when the dogs walked or trotted. In the contralateral forelimb, the peak force was unchanged, but the mean force significantly increased during walking and trotting; vertical impulse increased only during walking. Relative stance duration increased in the ipsilateral hind limb when the dogs trotted. In the contralateral fore and hind limbs, relative stance duration increased during walking and trotting, but decreased in the ipsilateral forelimb during walking. Analysis of load redistribution and temporal gait changes during hind limb lameness showed that compensatory mechanisms were similar regardless of gait. The centre of mass consistently shifted to the contralateral body side and cranio-caudally to the side opposite the affected limb. These biomechanical changes indicate substantial short- and long-term effects of hind limb lameness on the musculoskeletal system.

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Introduction

More than half of the musculoskeletal problems in dogs are caused by joint diseases affecting the hind limb (i.e. hip and knee; Johnson et al., 1994) and are commonly associated with alterations in the gait (i.e. lameness) due to the animal's effort to unload the affected limb. The biomechanical consequences of orthopaedic diseases, such as hip dysplasia and cranial cruciate ligament rupture, have been evaluated by investigating the load bearing characteristics of either the affected hind limb only (Gordon et al., 2003; Hoelzler et al., 2004; Conzemius et al., 2005; Evans et al., 2005; Madore et al., 2007) or both hind limbs (Budsberg et al., 1988; Voss et al., 2008; Ragetly et al., 2010; Bøddeker et al., 2012; Seibert et al., 2012). Fewer studies have evaluated the effects of hind limb lameness on the forelimbs (Rumph et al., 1993, 1995; Dupuis et al., 1994; Jevens et al., 1996; Katic et al., 2009). Nevertheless, unloading one limb causes biomechanical adaptations in all remaining limbs and results in an irregular gait pattern and a compensatory redistribution of limb loading.

During the standard orthopaedic examination, dogs are usually ambulated at different speeds and gaits to diagnose lameness. Since locomotor forces increase with speed and depend on gait (Riggs et al., 1993; McLaughlin and Roush, 1994; Renberg et al., 1999; Evans et al., 2003; Voss et al., 2010), not only may lameness be more apparent during trotting compared to walking (Quinn et al., 2007; Voss et al., 2007), but the compensatory mechanism used by the animal may differ. Additionally, the fundamental biomechanical differences between walking and trotting (i.e. legs behaving like inverted pendula vs. 'pogo sticks'; Cavagna et al., 1977) may result in differences in the locomotor adaptations to lameness. Previous studies have evaluated induced hind limb lameness in trotting dogs (O'Connor et al., 1989; Rumph et al., 1993, 1995; Dupuis et al., 1994) or clinical lameness in walking dogs (Budsberg et al., 1988; Katic et al., 2009; Bøddeker et al., 2012). It is uncertain if dogs show the same locomotor adaptations to hind limb lameness when walking and trotting.

In dogs, as in most quadrupedal mammals, fore and hind limbs play different functional roles during locomotion (Gray, 1968). The forelimbs exert a net-braking force, while the hind limbs exert a net-propulsive force during steady state locomotion (Budsberg et al., 1987; Riggs et al., 1993; Bertram et al., 1997; Lee et al., 1999). Forelimbs function as compliant struts (Carrier et al.,

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2008), whereas hind limbs function as levers (Schilling et al., 2009). Regardless of gait, the forelimbs bear a greater proportion of the dog's bodyweight (BW) in comparison with the hind limbs (Budsberg et al., 1987; Rumph et al., 1994; Bertram et al., 2000; Bockstahler et al., 2007; Voss et al., 2010). Therefore, when the function of a limb is partially lost, the dog's mechanism to cope with this loss has been expected to differ depending on whether a fore or a hind limb is affected (Roy, 1971; Leach et al., 1977).

To gain a better understanding of the compensatory load shifting mechanisms in lame dogs, we induced a moderate, reversible, load-bearing hind limb lameness in Beagles while they were walking and trotting on an instrumented treadmill, and evaluated the changes in the ground reaction force (GRF) and temporal gait parameters. Force plate analysis was used because it is an accurate and objective way of evaluating limb function and provides a reproducible measurement of the load bearing characteristics of the limbs. Inducing hind limb lameness allowed for a direct comparison of the gait parameters between sound and lame dogs and the precise determination of the degree and cause of lameness.

The aims of this study were: (1) to determine changes occurring in the vertical GRF, i.e. peak vertical forces (PFz), mean vertical forces (MFz) and vertical impulse (IFz), as well as the temporal gait parameters (i.e. footfall pattern, relative stance duration) in all four limbs; (2) to determine whether the observed locomotor adaptations differed between the gaits; and (3) to determine whether the compensatory mechanisms in response to hind limb lameness differed from the ones used to cope with forelimb lameness. To address the latter, we compared our results with the ones from a previous study (Abdelhadi et al., 2013), which used the same experimental design and therefore allowed for a direct comparison of load redistribution strategies.

Materials and methods

Animals

Eight Beagles aged 4 ± 1 years (mean $\pm$ standard deviation, SD) were used in this study. The sample size sufficient for this study was determined using Win Episcope 2.0 with a level of confidence of 95%, a power of 80% and the outcome measures PFz, MFz and IFz (Thrusfield et al., 2001). The BW (mean $\pm$ SD) of the two females and six males was 15.2 ± 1.1 kg. All dogs belonged to the Beagle population of the Small Animal Clinic of the University of Veterinary Medicine, Hannover, Germany. Inclusion criteria were absence of lameness (see results) and orthopaedic abnormalities in the previous clinical examination. Before data collection, the dogs were habituated to ambulating on the treadmill. Data collection started as soon as the dogs were walking and trotting smoothly and comfortably. All experiments were carried out in accordance with the German Animal Welfare guidelines of the State of Lower Saxony (approval number 12/0717).

Study design

To allow for comparison of the compensatory load redistribution mechanisms in forelimb vs. hind limb lameness, the same experimental protocol as in Abdelhadi et al. (2013) was used in the current study, but lameness was induced in the right hind limb (i.e. ipsilateral hind limb, H_i). Before inducing lameness, each Beagle walked (0.9 m/s) and trotted (1.4 m/s) on a horizontal treadmill. Despite the short temporal overlap in ground contacts between the forelimbs at the faster gait (duty factor, $D > 0.5$; i.e. running walk according to Hildebrand (1966); Fig. 1), from a mechanical point of view, this gait represented a trot (i.e. using spring-mass mechanics; Cavagna et al., 1977) and will subsequently be referred to as trot. The selected speeds were determined during the habituation sessions, as were the preferred speeds for the dogs at the respective gait. At these speeds, the dogs ambulated smoothly and comfortably matched the treadmill speed, which allowed us to record single limb forces (see below).

After recording the control trials, a moderate and reversible hind limb lameness was induced using a small sphere, which was coated with cotton gauze and taped under the paw with adhesive tape and bandages. The size of the sphere (9.5 or 16.0 mm in diameter) depended on the degree of lameness it induced in a given dog. The degree of lameness was evaluated based on the GRF data. To be able to compare our data with previous results (Abdelhadi et al., 2013), we aimed at an unloading of ~30–40% with regard to PFz (in %BW) compared with the sound condition.

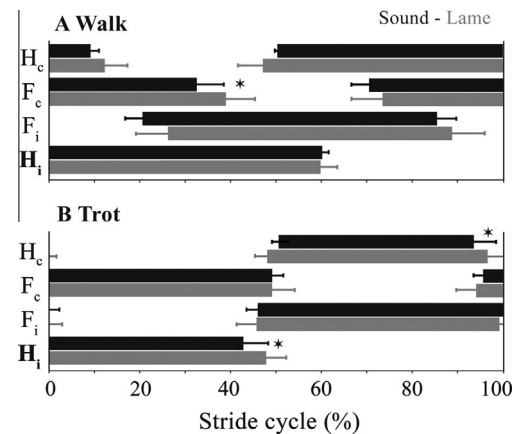


Fig. 1. Stride cycle with mean $\pm$ standard deviation (SD) stance duration for footfall patterns of all eight Beagles during walking (A) and trotting (B) for the sound (control) condition (black bars) and after lameness was induced (grey bars) in the right hind limb (ipsilateral hind limb, H_i). Stance duration was the interval between touch-down to lift-off of a paw. *Touch-down of the paw of a limb after induced lameness was shifted significantly during walking (F_c , $P < 0.05$) and trotting (H_c , $P < 0.05$; H_i , $P < 0.01$) in comparison with the results for that same limb in the sound condition. Stride cycle was the interval between touch-down to next the touch-down of the same limb. F_c , contralateral forelimb; F_i , ipsilateral forelimb; H_c , contralateral hind limb; H_i , ipsilateral hind limb (i.e. affected limb, bold).

Data collection and analysis

Data collection and analysis are described in detail in Abdelhadi et al. (2013). A treadmill with four separate belts and force plates underneath each belt (Model 4060-08, Bertec) was used to record single limb GRF (sampling rate 1000 Hz). Data were recorded and evaluated to ensure a sufficient number of valid steps using Vicon Nexus (Vicon). Control data comprising at least 5–10 trials, each lasting up to 30 s and covering between 48 and 65 strides, were recorded for each dog while walking and trotting comfortably at the selected speed. After a break of approximately 15 min, lameness was induced and the data collection repeated. To evaluate kinetic changes, 10 valid consecutive strides were selected for each dog, gait and condition.

Mean $\pm$ SD for PFz, MFz and IFz were calculated for all four limbs. Additionally, relative stance duration (i.e. duration of stance phase as percentage of total stride duration = D), as well as symmetry indices for the vertical forces of the fore and hind limbs, were determined. After manual identification of the footfall events in Vicon Nexus using the GRF, the force data were time-normalised to 100% of the stance duration of the respective limb and transferred to Microsoft Excel for further analysis. The vertical force parameters were then normalised to the dog's BW using the following equation:

$$\text{GRFs (\%BW)} = \text{Fz} \times 100 / (\text{BM} \times 9.81).$$

These BW-normalised data were used to compare the load-bearing characteristics among the four limbs before and after lameness was induced (Steiss et al., 1982):

$$\% \text{BW bearing} = \text{Fz of the limb} / \text{total Fz of all limbs} \times 100.$$

Symmetry in the vertical force and temporal variables was quantified using the following equation (Herzog et al., 1989):

$$\text{SI} = 100 \times (X_i - X_c) / (0.5 \times [X_i + X_c]).$$

In this equation, X represents the mean value of PFz, MFz or IFz of the ipsilateral (i) and the contralateral (c) limbs from the 10 steps. Footfall patterns and D were evaluated to test for significant differences in the temporal gait parameters due to lameness. A stride cycle begins with the contact of the affected limb (ipsilateral hind limb, H_i) and ends with its subsequent touch-down. Therefore, one locomotor cycle comprises one complete stance and one complete swing phase of the reference limb.

Statistical analysis

Data were tested for normal distribution using the Kolmogorov–Smirnov test. The significance of the differences in PFz, MFz and IFz between the sound and lame conditions was determined using one-way analysis of variance (ANOVA) for repeated measures, followed by a post hoc Tukey test. Paired t tests were used to compare relative stance durations between sound and lame conditions. P values < 0.05 were considered to be significant. All statistical tests were performed in GraphPad Prism (version 4).

Results

Load distribution

No significant differences in GRF (i.e. PFz, MFz and IFz) between the two forelimbs and between the two hind limbs were observed during walking and during trotting before lameness was induced (Table 1). Compared to the sound condition, all parameters decreased significantly in the ipsilateral hind limb when lameness was induced. During walking, these changes were primarily due to a decrease of the first peak in the m-shaped force curve (Fig. 2). In the contralateral hind limb, PFz, MFz and IFz increased significantly at both gaits. Additionally, PFz increased significantly during walking and trotting in the ipsilateral forelimb, while no significant changes were observed in MFz and IFz. While PFz remained unchanged, MFz significantly increased during both walking and trotting in the contralateral forelimb. In this limb, IFz increased during walking, but not during trotting (Fig. 3).

Symmetry indices

For the sound condition, no asymmetry was detected between the right and left forelimb, or between the right and left hind limb, either during trotting or during walking (Table 2). After lameness was induced, asymmetry significantly increased in all GRF parameters of the hind limbs during walking and trotting. Symmetry indices for PFz and IFz for the forelimbs indicated significant changes for walking but not for trotting.

Relative stance duration

In the lame condition, relative stance duration significantly increased in the ipsilateral hind limb while the dogs were trotting; thereby lift-off was significantly delayed (Table 3; Fig. 1). Relative stance duration was significantly increased in the contralateral forelimb and hind limb during walking and trotting compared to the sound condition. This increased stance duration was associated with a significant later lift-off of the contralateral forelimb during

walking and of the contralateral hind limb during trotting. Relative stance duration significantly decreased in the ipsilateral forelimb during walking.

Discussion

Prior to the induction of lameness, we observed no significant differences in GRF values (PFz, MFz and IFz) between the two forelimbs and the two hind limbs in walking and trotting dogs. Correspondingly, the calculated symmetry indices were in the normal range (<6%, Budsberg et al., 1993; <3.2%, Fanchon and Grandjean, 2007), which indicated symmetrical load distribution between the two forelimbs and the two hind limbs. The values of this study compared well with previous observations in dogs (Budsberg et al., 1993; Fanchon and Grandjean, 2007; Voss et al., 2008). Each forelimb bore about 30% and each hind limb about 20% of the dog's BW. Comparison with previous results (Budsberg et al., 1987; Jevens et al., 1993; Rumph et al., 1994; Bertram et al., 2000; Bockstahler et al., 2007) confirmed that the Beagles in the current study were sound and affirmed that the data collected before hind limb lameness was induced were valid as a control.

In most published studies in which hind limb lameness was induced in dogs, the vertical GRF parameters (PFz, MFz and IFz) were significantly lower in the affected limb (Fig. 3) and IFz and PFz were increased in the contralateral hind limb (O'Connor et al., 1989; Rumph et al., 1993, 1995; Dupuis et al., 1994; Jevens et al., 1996), the exceptions to these studies being Budsberg (2001) and Ballagas et al. (2004). However, adaptations in the vertical GRF parameters in forelimbs varied among studies (Fig. 3). Similar to the observations in induced lameness, most clinical studies observed a decrease in GRF values in the affected hind limb and an increase in these values in the contralateral hind limb, while the results for the forelimbs varied again among the studies (Fig. 3). Taken together, comparison of the results from this study with clinical studies shows that the load-redistribution pattern observed in walking dogs which were lame due to stifle disease was most comparable with the results from this study.

A confounding factor when comparing different studies is that one or more parameters critical to the recorded gait parameters may vary. For example, speed, gait or breeds have been shown to affect GRF characteristics (Riggs et al., 1993; McLaughlin and Roush, 1994; Renberg et al., 1999; Evans et al., 2003; Mölsa et al., 2010; Voss et al., 2010, 2011). Furthermore, differences in data collection (e.g. force plates in a walkway vs. instrumented treadmill) or the degree and cause of lameness (e.g. location within the limb or induced vs. clinical lameness) may have an impact on the compensatory load shifting. For example, unloading of the affected limb in dogs with induced stifle joint lameness due to transection of the cranial cruciate ligament or synovitis (33–81% reduction in PFz; O'Connor et al., 1989; Rumph et al., 1993, 1995; Dupuis et al., 1994; Jevens et al., 1996; Budsberg, 2001; Ballagas et al., 2004) generally produced a greater effect than in dogs with clinical stifle or hip joint lameness (2–18%; Budsberg et al., 1988; Hofmann, 2002; Katic et al., 2009; Bøddeker et al., 2012). This may partially explain why the locomotor adaptations to hind limb lameness observed in this study, inducing a 34 ± 9% reduction in PFz, compared better with the more severe lameness studied previously (Dupuis et al., 1994).

To limit variability in the results introduced by differences among patient populations and the experimental setting, and to allow for the direct comparison between the sound and the lame condition in the same individual, lameness was induced in this study instead of enrolling patients. Furthermore, the induced lameness model controlled for the cause and degree of lameness. Therefore, in contrast to clinical patients, which may only show lameness

Table 1

Load distribution indicated by peak vertical forces (PFz), mean vertical forces (MFz) and vertical impulse (IFz) in % bodyweight (BW) for the sound condition and when lameness was induced in the right hind limb (ipsilateral hind limb, H_i).

	Sound		Lame			Sound		Lame	
	F_c		F_i			F_c		F_i	
Walk									
PFz	29.2 ± 0.8		30.9 ± 1.2NS			29.3 ± 0.8		32.0 ± 0.7***	
MFz	30.5 ± 0.7		32.6 ± 1.0*			29.8 ± 1.0		30.9 ± 0.8NS	
IFz	31.3 ± 1.0		33.6 ± 1.3*			30.6 ± 1.1		30.4 ± 1.1NS	
Trot									
PFz	29.6 ± 1.2		31.0 ± 1.5NS			29.8 ± 1.1		31.9 ± 1.1*	
MFz	29.0 ± 1.1		31.1 ± 1.6*			28.5 ± 1.2		30.2 ± 1.4NS	
IFz	31.4 ± 1.5		33.2 ± 2.0NS			31.1 ± 1.2		31.2 ± 1.6NS	
	H_c		H_i			H_c		H_i	
Walk									
PFz	20.8 ± 0.9		23.6 ± 1.3***			20.7 ± 0.7		13.5 ± 1.8***	
MFz	19.8 ± 0.9		22.2 ± 1.5**			19.9 ± 0.8		14.4 ± 2.1***	
IFz	18.9 ± 1.2		22.3 ± 1.5***			19.2 ± 0.7		13.7 ± 2.1***	
Trot									
PFz	20.4 ± 1.0		23.6 ± 0.8***			20.3 ± 1.8		13.5 ± 1.6***	
MFz	21.3 ± 1.1		24.5 ± 1.0***			21.3 ± 1.3		14.2 ± 1.7***	
IFz	18.9 ± 1.2		22.5 ± 1.6***			18.6 ± 1.5		13.1 ± 1.9***	

NS, not significant.

* Significant at $P < 0.05$.

** Significant at $P < 0.01$.

*** Significant at $P < 0.001$.

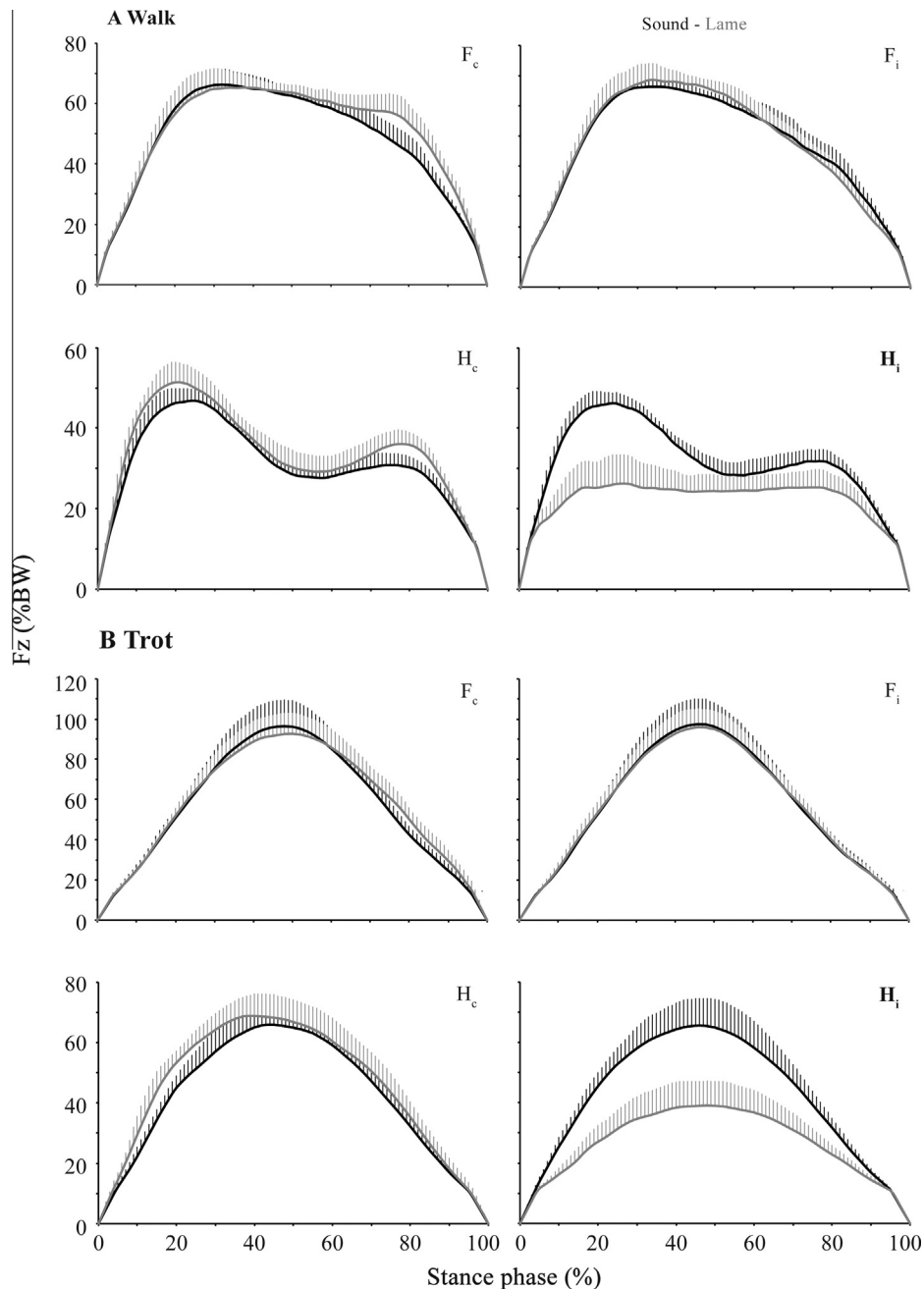


Fig. 2. Vertical ground reaction forces of all eight Beagles during walking (A) and trotting (B) for the sound (control) condition (black) and after lameness was induced (grey) in the right hind limb (ipsilateral hind limb, H_i). Plotted are the mean values, with the standard deviations (SDs) as error bars. Stance duration was the interval between touch-down to lift-off of the respective limb. F_c , contralateral forelimb; F_i , ipsilateral forelimb; H_c , contralateral hind limb; H_i , ipsilateral hind limb (i.e. affected limb, bold).

during faster gaits when greater locomotor forces act and thus the level of discomfort and pain increases, the degree of unloading was the same during walking and trotting in this study, forcing the animal to redistribute loading for both gaits.

Despite profound mechanical differences between the two gaits studied, such as the inverted pendulum vs. the spring-mass behaviour of the legs and thus differences in the trajectory of the centre of mass (CoM) of the body (Dickinson et al., 2000), the load redistribution was comparable in this study. In both gaits, the BW was shifted to the contralateral side and cranially. Only one difference was observed; IFz increased in the contralateral forelimb during walking, but did not significantly change during trotting (Fig. 3). Similarly, in dogs with induced forelimb lameness, load redistribution was comparable between walking and trotting (except PFz in

the contralateral forelimb; Abdelhadi et al., 2013). Therefore, all other factors being equal (e.g. cause and degree of unloading, breed), dogs which walk and trot at their preferred speeds appear to redistribute the loading of the limbs in a similar way.

Furthermore, during both walking and trotting, relative stance duration increased in both limbs contralateral to the affected limb. However, gait-dependent differences were observed in the stance adaptations of the ipsilateral limbs; relative stance duration decreased in the forelimb during walking and increased in the hind limb during trotting. The shorter contact time, associated with a larger peak force, most likely acted as a compensatory mechanism so that the mean force and impulse in the ipsilateral forelimb could be maintained. The longer contact times of both hind limbs during trotting resulted in a reduction of the shared swing phase of the

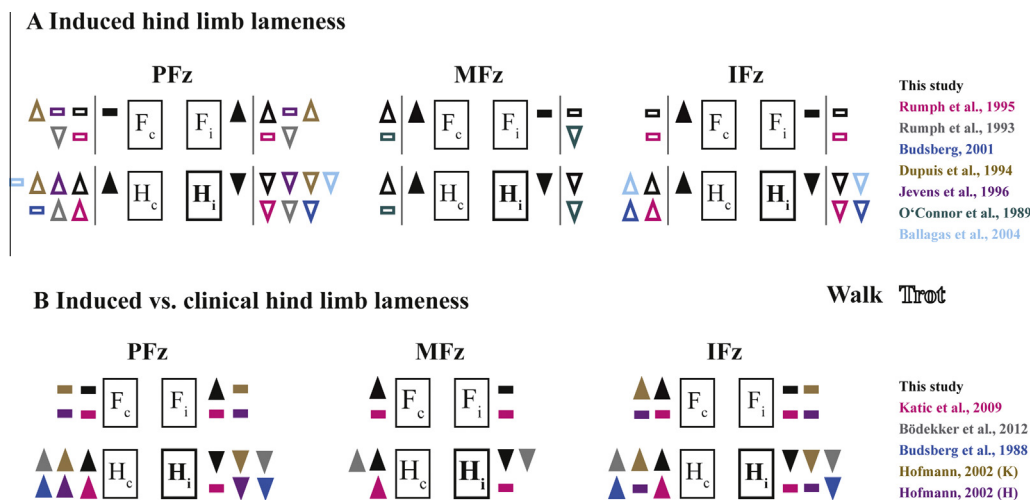


Fig. 3. Summary of the results of this study and previous studies illustrating the changes in the GRF parameters due to induced (A) and clinical (B) hind limb lameness. Triangles pointing upwards indicate a significant increase in the respective parameter and limb. Triangles pointing downwards indicate a significant decrease of the parameter. Horizontal bars represent no change. Solid symbols illustrate the observations for the walk, open symbols for the trot. F_c , contralateral forelimb; F_i , ipsilateral forelimb; H_c , contralateral hind limb; H_i , ipsilateral hind limb (i.e. affected limb, bold); PFz, peak vertical force; MFz, mean vertical force; IFz, vertical impulse.

Table 2

Symmetry indices for the sound condition and when lameness was induced in the right hind limb (i.e. ipsilateral hind limb).

	Forelimb		Hind limb	
	Sound	Lame	Sound	Lame
<i>Walk</i>				
PFz	0.3 ± 1.6	3.6 ± 2.6**	-0.5 ± 1.7	-54.8 ± 14.9***
MFz	-2.5 ± 2.8	-5.3 ± 3.1NS	0.8 ± 4.0	-42.7 ± 19.3***
IFz	-2.2 ± 3.7	-10.2 ± 5.1*	1.9 ± 5.5	-47.9 ± 19.3***
<i>Trot</i>				
PFz	0.8 ± 1.6	3.1 ± 7.1NS	-0.7 ± 1.6	-55.1 ± 13.2***
MFz	-1.8 ± 2.3	-3.1 ± 8.2NS	0.0 ± 3.9	-53.6 ± 13.0***
IFz	-1.1 ± 2.3	-6.2 ± 9.0NS	-1.9 ± 3.9	-53.7 ± 16.6***

Perfect symmetry is given at SI = 0. Negative values indicate that the respective parameter was greater for the contralateral than the ipsilateral limb; positive values indicate the opposite.

NS, not significant.

* Significant at $P < 0.05$.

** Significant at $P < 0.01$.

*** Significant at $P < 0.001$.

Table 3

Relative stance duration (D) in sound dogs and when lameness was induced in the right hind limb (i.e. ipsilateral hind limb, H_i).

F_c		F_i	
Sound	Lame	Sound	Lame
<i>Walk</i>			
64.0 ± 1.0	65.7 ± 1.4*	64.1 ± 1.5	62.2 ± 2.0**
<i>Trot</i>			
53.6 ± 2.7	55.8 ± 1.9**	54.0 ± 2.4	54.2 ± 2.4NS
H_c		H_i	
<i>Walk</i>			
59.2 ± 1.6	64.8 ± 2.5***	60.2 ± 1.5	59.3 ± 3.7NS
<i>Trot</i>			
44.0 ± 4.4	48.6 ± 4.9**	43.9 ± 4.4	48.2 ± 4.0**

NS, not significant.

* Significant at $P < 0.05$.

** Significant at $P < 0.01$.

*** Significant at $P < 0.001$.

hind limbs. During walking, the significant increase in the relative stance duration of the contralateral forelimb was first and foremost due to a later lift-off. Since this later lift-off is associated with a greater limb retroversion (S. Fischer unpublished observations), the paw of the contralateral forelimb would be placed closer to the CoM, which may facilitate the unloading of the affected hind limb (Roy, 1971). Compared to the sound condition, relative stance duration of the affected hind limb decreased significantly during walking, but not during trotting.

In dogs, the cranio-caudal body mass distribution has been shown to vary slightly among breeds, but the forelimbs consistently bear a greater percentage of BW than the hind limbs. In the Beagles, as in other mesomorphic breeds, the forelimbs bear ~60% and the hind limbs ~40% of the BW (Rumph et al., 1994; Bertram et al., 2000; Katic et al., 2009; Abdelhadi et al., 2013). Due to this difference in loading, the dog's mechanism to cope with fore vs. hind limb lameness was expected to differ (O'Connor et al., 1989; Rumph et al., 1993, 1995; Dupuis et al., 1994; Jevens et al., 1996; Katic et al., 2009). However, the comparison of the changes in the vertical GRF values and the temporal gait parameters shows some striking similarities.

Since the same experimental design, cause and degree of lameness were used, the results of the present study are comparable with those of Abdelhadi et al. (2013); in both studies, there were reductions in PFz at walk ($35 \pm 9\%$ vs. $34 \pm 4\%$, respectively) and trot ($33 \pm 9\%$ vs. $34 \pm 12\%$), respectively. While the ipsilateral hind limb showed no change when forelimb lameness was induced, PFz increased in the ipsilateral forelimb in hind limb lameness. However, this increase was combined with a decreased stance duration, which resulted in unchanged MFz and IFz, similar to the results for the ipsilateral hind limb in forelimb lameness (Abdelhadi et al., 2013).

Thus, changes in load redistribution and temporal gait variables were very similar and appeared not to be dependent on whether a forelimb or a hind limb was affected. These observations suggest that: (1) ground contact times decreased in the limb ipsilateral to the affected limb and increased in the contralateral limbs; and (2) the CoM was shifted to the contralateral body side and to the rear in forelimb lameness and to the front in hind limb lameness.

Conclusions

The lameness model used in this study controlled several parameters, which probably introduce variability in the results and permitted precise determination of the degree and cause of lameness. When all variables are comparable, dogs ambulating at their preferred speeds redistribute the load among the limbs and adapted their temporal gait parameters regardless of the gait and whether a forelimb or a hind limb is affected. Dogs unloaded the affected limb and shifted the CoM to the contralateral side and cranio-caudally to the side opposite to the affected limb.

Conflict of interest statement

None of the authors has any financial or personal relationships which could inappropriately influence or bias the content of the paper.

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